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Jitter Tool : Running Jitter Tool

Running Jitter Tool

To run Jitter Tool, you must first call a function that exports a file from a DSP function that has analyzed the sample jitter measurement data. To save the results of a jitter test to a file, use this VBT language in a DSP function:

- SB6G/UltraSerial10G: TheHdw.Serial.Jitter.FileExport.
- UltraPin1600: TheHdw.Digital.Jitter.FileExport.

These methods export jitter and eye files containing DSP wave and setup information to a specified file name for Jitter Tool use.

The following example exports file data for each of the digital jitter measurements. The source DSP wave in the DSP function must be marked as ByRef for data to be brought up to the host computer correctly. This will forcibly wait for the DSP procedure to complete.

```
If (FileExportData = True) Then
   Dim theWave As dspWave
   For Each thePin In dspWaves.pins
      For Each theWave In thePin.Value
         Call TheHdw.Digital.Jitter.FileExport(theWave, _
            "C:\temp\jitterFile_" & thePin.Name & "_" & theWave.Site)
      Next theWave
   Next thePin
End If
```

If the DSP wave object is associated with an eye test (having multiple VOD levels worth of data), a file is produced with eye data, per-edge jitter data for the main VOD, raw data, and the setup parameters.

If the DSP wave object is associated with a jitter test, a file is produced with per-edge jitter data, raw data, and the setup parameters.

If jitter analysis fails, the file export still outputs the raw data and setup parameters to the file.

Once you have exported a file that contains the results of a jitter test, launch jitter tool from the Start menu: Start>Programs>Teradyne IG-XL Vx.xx.xx_uflx>Tools>Jitter Tool. Then open the file with Jitter Tool to see the results.

☑ **Note:**

You can run multiple instances of Jitter Tool on a single computer as long as memory constraints allow this.

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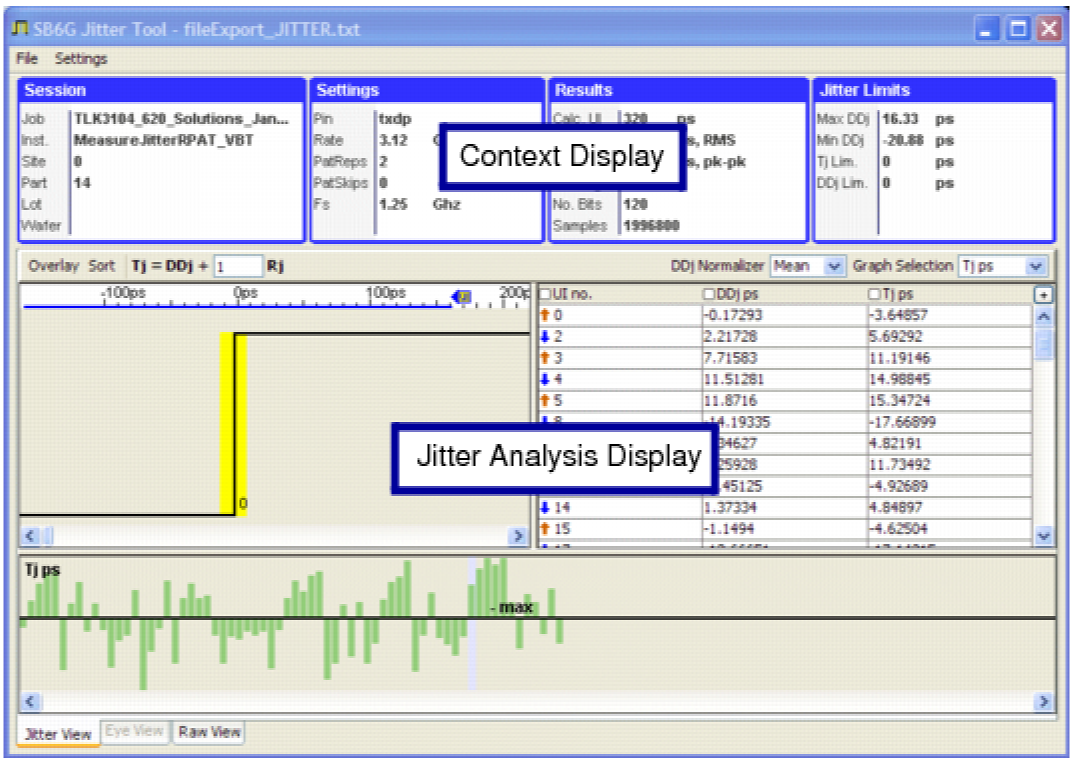
Jitter Tool : Main Display

Main Display

The following figure shows the main Jitter Tool display.



Jitter Tool Window

The primary display of Jitter Tool has these sections:

Context Display	Displays the conditions under which the jitter data was collected. The context display has five possible panes. Which pane is visible depends on whether jitter data, eye data, or raw data is loaded.
Jitter Analysis Display	Displays one of three views:

- Jitter View displays the jitter results on a per edge basis.
- Eye View displays the rolled up data for one or more voltage levels.
- Raw View shows the in-order sampled data.

These displays are managed inside a tabbed pane, making them mutually exclusive.

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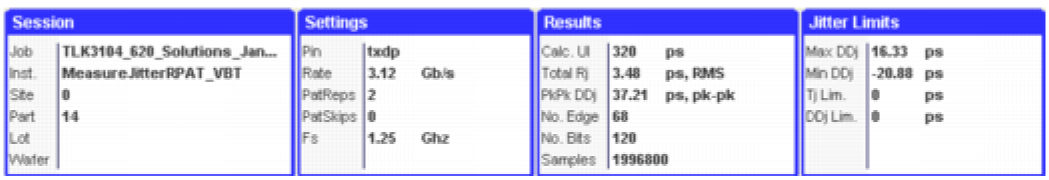
Jitter Tool : Context Display

Context Display

The context display can show up to five panes, depending on what kind of data is loaded in the tool. You can copy the information in the context display to the clipboard by right-clicking the display and selecting Copy Information to Clipboard.



Jitter Tool Context Display

The following values are shown in the panes of the context display:

Session	Displays the test program (job) name, test instance name, Site ID, Part ID, Lot ID, and Wafer ID. This pane is shown with jitter, eye, or raw data loaded.
Settings	Shows the Pin, Data Rate, Pattern Repetitions, Number of Pattern Skips, and Walking Strobe Frequency. This pane is shown with jitter, eye, or raw data loaded.
Results	Shows the calculated unit interval, the rolled-up Rj and DDj, the number of bits and edges, and the effective Sampling Frequency. This pane is shown with jitter or eye data loaded.
Jitter Limits	Shows jitter limits. The maximum and minimum DDj values are shown in this pane. Also shown are DDj and Tj limits. This pane is shown with jitter or eye data loaded.

Eye Limits	Shows eye limits. This pane is shown when eye data is loaded. These values are special in that they do not come from the file, but are values you specify. They appear as links in the context display. Clicking them brings up an editor that allows you to set the values.

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Jitter Tool : Jitter Analysis Display

Jitter Analysis Display

You can only view the three views of the Jitter Analysis Display (Jitter View, Eye View, and Raw View) one at a time. Note that you can export and view UltraPin1600 pulse width data using the Jitter View because that data is generated using a two-bit jitter pattern.

This section describes each view in turn.

- Jitter View
- Eye View
- Raw View

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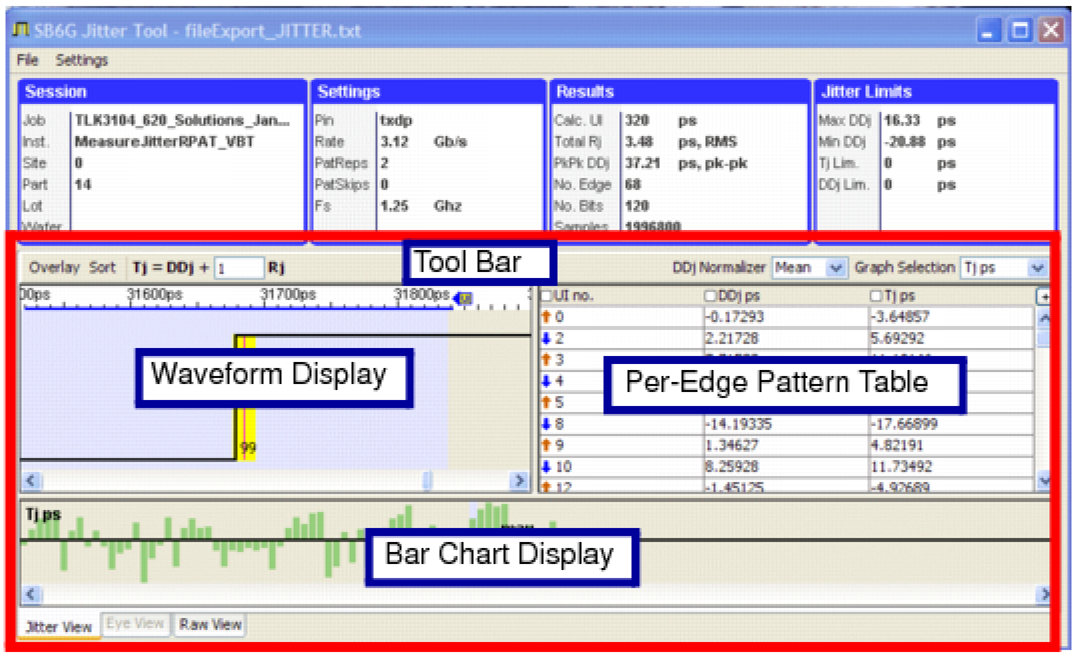
Jitter Tool : Jitter Analysis Display : Jitter View

Jitter View



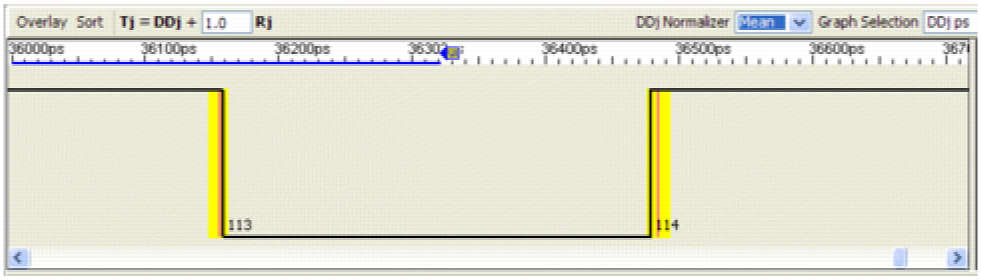
Jitter Analysis Display: Per-edge Jitter View

The Jitter View has the following parts:

•	Waveform Display
•	Bar Chart Display
•	Toolbar Buttons
•	Per-Edge Pattern Data Table

Waveform Display

The Waveform display shows the idealized waveform along with the jitter values for each edge.



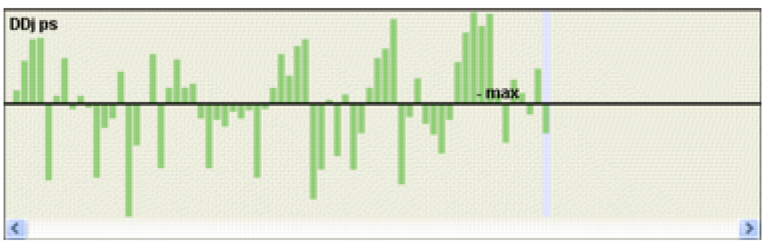
Waveform Display

The display shows the rising and falling edges and the corresponding cycle numbers. The red line shows the DDj for the edge and the yellow bar represents the min and the max edge values. All values are displayed in picoseconds. To change the scale to which the waveform is drawn, drag the blue line shown on the x-axis (above the waveform).

You can copy this image to the clipboard by right clicking the image and selecting Copy View to Clipboard.

Bar Chart Display

You can configure the bar chart display to display a bar graph for values in any column in the per-edge pattern table.



Bar Chart Display

The figure shows a bar chart for DDj values. The thick horizontal black line is the base line. The bars above the base line indicate positive DDj values, and the bars below the baseline represent negative DDj values. The display highlights the highest value/tallest bar in the currently visible portion of the display and labels it max. This identification will change as you scroll through the display. You can copy this image to the clipboard by right clicking the image and selecting Copy View to Clipboard.

Toolbar Buttons

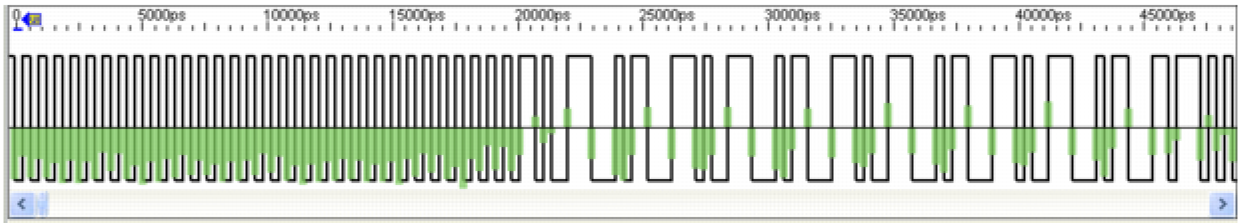
The Graph Selection drop-down box on the tool bar selects which per-edge value is shown in the bar chart display. In the example shown above, DDj ps is shown. The options are:

Edge no.	Edge number for each edge.
UI no.	Unit interval for each edge.
Run Len	Run length for each edge.

DDj ps	Deterministic data jitter for each edge.
Tj ps	Total jitter for each edge in ps.
Early ps	Early value for each edge in ps.
Late ps	Late value for each edge in ps.

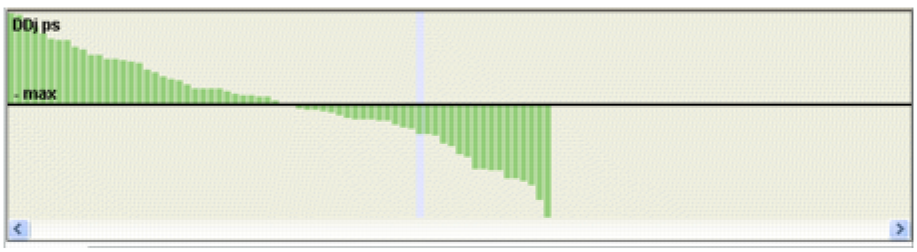
If overlay mode is on (explained below), changing this value also affects the waveform panel.

The Overlay button, located on the tool bar, toggles the waveform display. It switches the display between waveform showing jitter values to waveform overlaid with the bar graph data.



Waveform Bar Graph Overlay

The Sort button, located on the tool bar, toggles the bar chart display. The sort display reorders the data in descending order. Compare this figure with 1-5.

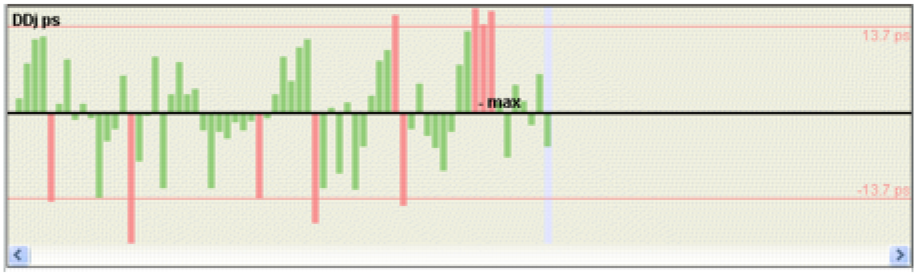


Bar Chart Display with Sort Button Selected

The Tj equation on the tool bar modifies the definition of Tj through specifying the Rj multiplier. Changing the Rj multiplier value affects the Tj value shown in the data table, the bar graph panel if Tj is being displayed, and the waveform panel if Tj is being graphed and the overlay mode is selected. Note that the Rj multiplier value is for scaling purposes. For positive values of DDj the Rj multiplier value is additive to increase the total positive Tj value. For negative values of DDj the Rj multiplier is subtractive to increase the total negative value of Tj.

The DDj Normalizer box on the tool bar selects either a Mean or Center methodology for normalizing the DDj values. Changing this mode affects not only the DDj values in the per-edge pattern data table, but also Tj value since this value is determined as a function of DDj. DDj and Tj values shown in the waveform display and bar chart display are also affected when this mode changes.

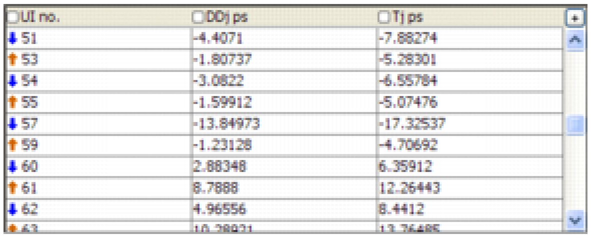
When limits for Tj or DDj have been specified in the Tool Configuration Menu dialog and Tj or DDj is displayed in the bar graph, edges that violate the limits are colored in red.



Bar Chart Display with Limits Set

Per-Edge Pattern Data Table

The pattern data table shows the per-edge jitter values in a tabular form.



UI no.	DDj ps	Tj ps
51	-4.4071	-7.88274
53	-1.80737	-5.28301
54	-3.0822	-6.55784
55	-1.59912	-5.07476
57	-13.84973	-17.32537
59	-1.23128	-4.70692
60	2.88348	6.35912
61	8.7888	12.26443
62	4.96556	8.4412

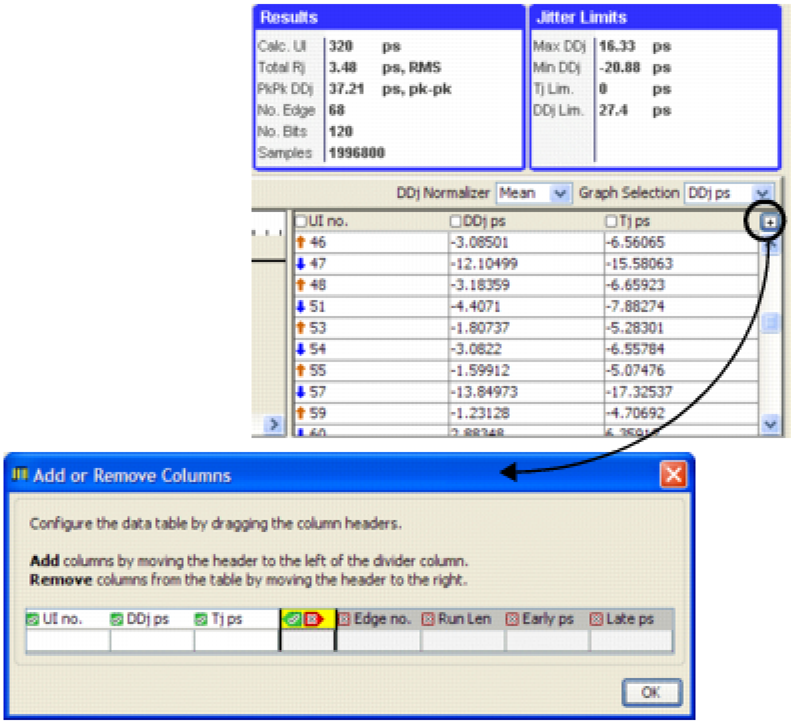
Per-Edge Pattern Data Table

You can sort the table on any column by selecting the box next to the title found in the table header. You can sort the table in ascending or descending order for any given combination of columns.

You can configure the table to display (or not display) any of the following columns:

- Unit Interval
- DDj in ps
- Tj in ps
- Edge Number
- Early value in ps
- Late value in ps
- Run Length

To configure the display, add or remove columns by clicking the + icon shown above the vertical scrollbar and bringing up the Add or Remove Columns dialog box as shown in the figure.



Adding or Removing Pattern Table Columns

You can add columns by dragging the column headers to the left of the divider and delete them by moving them to the right of the divider column. The Per-edge Pattern Table allows you to copy some or all rows in the data table to a clipboard. Once on the clipboard, you can post the information to an Excel spreadsheet, PowerPoint presentation, or any application that is clipboard aware. Place the mouse pointer over the table and right click to:

- Copy Selected Rows to Clipboard
- Copy Everything to Clipboard
- Clear Selection

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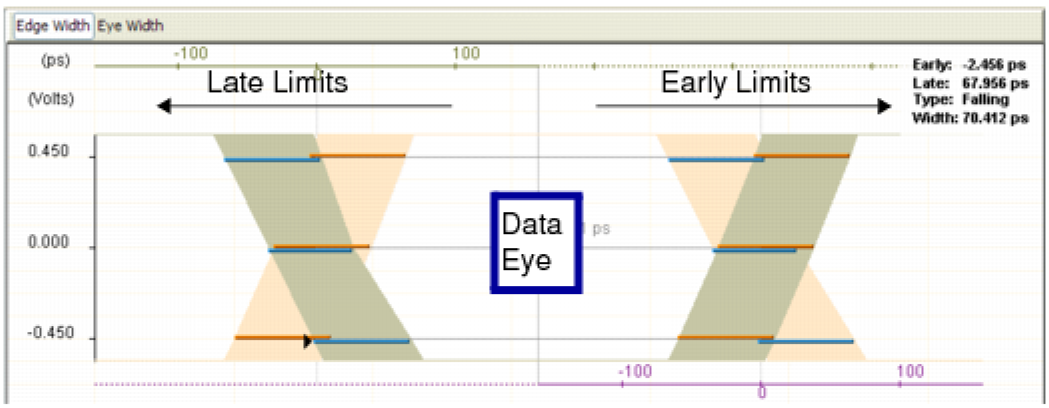
Jitter Tool : Jitter Analysis Display : Eye View

Eye View

The Eye Display shows the folded jitter values for one or more voltage levels. The x-axis shows the time scale in picoseconds. The range of the x axis is from negative UI/2 to positive UI/2. The y-axis shows the actual voltage at which the jitter measurement was taken. The y-axis is not drawn to scale to keep the diagram as symmetrical as possible.



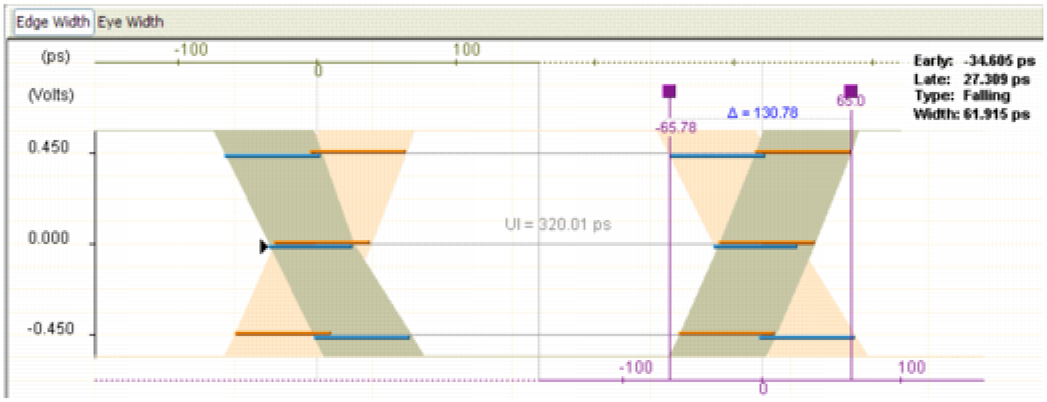
Jitter Analysis Display: Eye Display, Edge Width Selected

You can copy the eye diagram to the clipboard by right clicking the image and selecting Copy Eye Diagram to Clipboard.

Edge Width Mode

The tool bar in the eye view switches the eye view from Edge Width to Eye Width. When Edge Width is selected, an edge-centric view of the eye is shown. The folded jitter values for rising and falling edges are color coded to make them easily distinguishable. Bars in brown show the averages and the worst-case (earliest-earliest and latest-latest) values for the rising edges. Bars in blue show the averages and the worst-case values for the falling edges. Placing the mouse pointer over a bar causes late, early, and width statistics to appear in the upper-right corner of the display.

You can use markers on the display to find the difference in time between any two points in the figure.



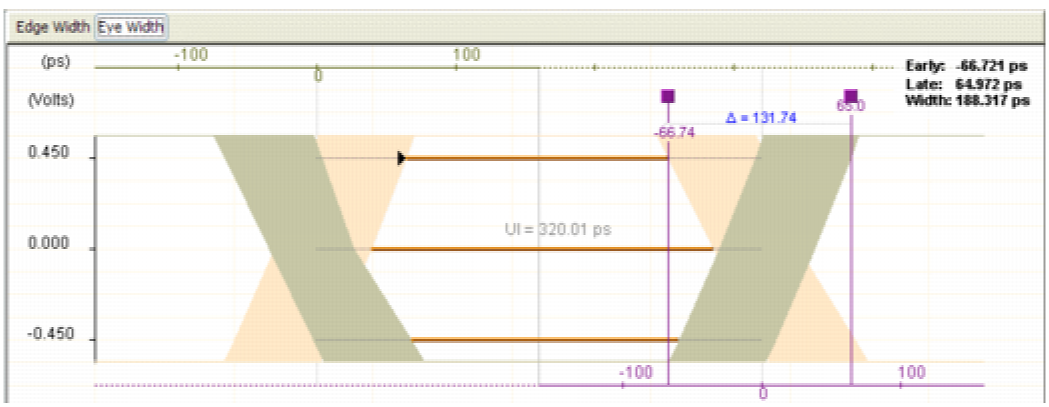
Using Markers with Eye Diagrams

The eye diagram can show up to two markers at any time. Invoke a marker by clicking the eye diagram. Use the knob on the marker (the square box) to drag the marker. Click the knob to delete the marker.

Eye Width Mode

When the Eye Width button is selected from the tool bar, an eye-centric mode is displayed. In the eye-centric mode, the width of the eye is drawn taking both rising and falling timing. Placing the mouse pointer over the brown bars shows the late, early, and width values for the eye in the upper-right corner of the display.

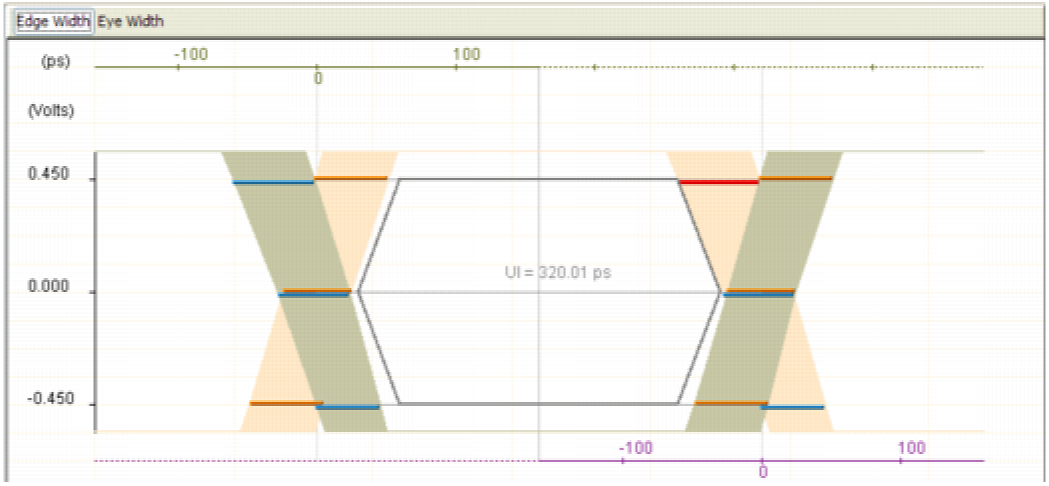
You can use markers with this Eye diagram as well. They operate the same as they do in the Edge Width mode.



Jitter Analysis Display: Eye Display, Eye Width Selected

Eye Template

When eye limits have been specified in the Tool Configuration Menu, the eye template will be drawn. The geometric shape in the center (a hexagon, in case of a 6-point eye) shows the template values, or limits at each voltage level. The points on the left side mark the latest limits, and the ones on the right side are the earliest limits. In edge-centric mode, any data bar that extends in to the eye template will be colored red to indicate failure. In eye-centric mode, any bar that does not extend outside the template is colored red.



Eye Display with Eye Template Drawn

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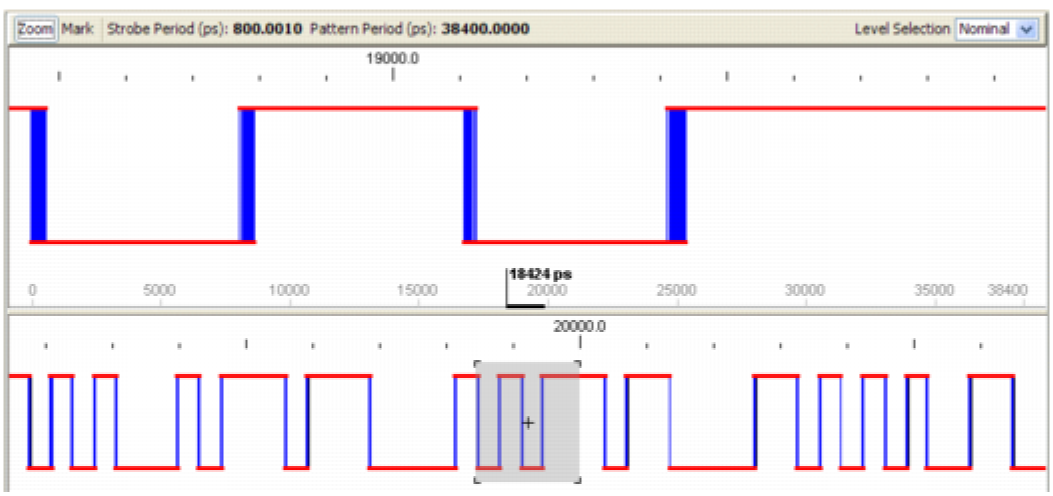
Jitter Tool : Jitter Analysis Display : Raw View

Raw View

The raw display reorders the out-of-order samples and plots those samples, drawing a waveform through the points. The main display is split into two views of the waveform. The upper view is zoomed in to focus on whatever you want to see. The lower panel is zoomed out, giving a sense of context.



Jitter Analysis Display, Raw Display View

When the mouse hovers over a red data sample in the upper panel (any point along any red line), information about that sample is displayed (effective time, real time, sample number, and pattern iteration).

Left-click the lower panel of this display to center both the upper and lower panel on that part of the waveform.

Zoom Mode

When the Zoom button is selected, the view is set to zoom mode. This mode is mutually exclusive with the marker mode described below. In zoom mode, left-clicking the upper panel zooms in and centers the display on the area of the waveform selected. Right-clicking the upper panel zooms out and centers the display on the area of the waveform selected.

☑ **Note:**

You can use the mouse wheel to zoom in and out regardless of whether zoom or marker mode is selected.

Marker Mode

When the Mark button is selected, the view is set to marker mode. In marker mode, left-clicking the upper panel places the first half of a marker. A subsequent left-click places the second half of the marker. If a right-click occurs instead, the initial half of the marker is removed. Once a marker is set, left-clicking the (round-shaped) knob of the marker and dragging changes the marker's location.

Right-clicking the knob of a marker brings up a menu. Using this menu, you can delete the marker or change its color.

When the mouse hovers over a marker, both marker locations and the delta between them are displayed.

VOD Levels

Use the menu on the tool bar to select which of the three different VOD levels is to be shown in the raw display. This menu is only operational when a file with eye data is loaded in Jitter Tool.

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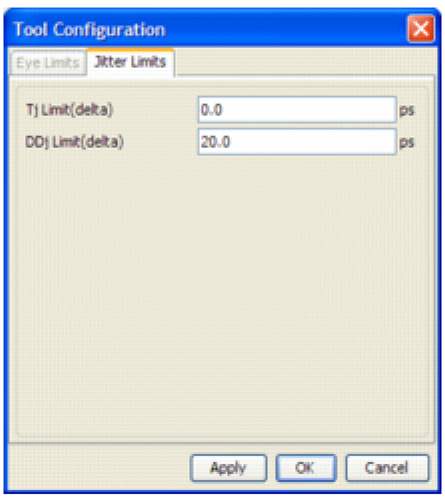
Jitter Tool : Tool Configuration Menu

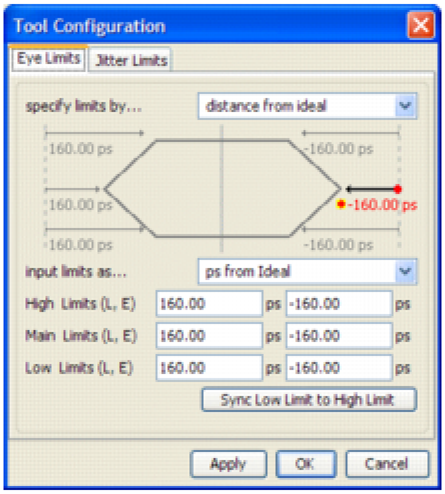
Tool Configuration Menu

From the Settings menu item at the top of the Jitter Tool window, you can set the Eye Limits or Jitter Limits. Eye limits can only be set if eye data is loaded. Likewise, jitter limits can only be set if jitter data is loaded.



Tool Configuration: Jitter Limits

The Tj Limit (delta) and DDj Limit (delta) fields specify desired jitter limits to be applied to the jitter display. When you click the Apply or OK button, the limits are drawn in the bar graph, and any failing bars (edges) are colored red. In overlay mode, the bars are red in the waveform display as well.



Tool Configuration, Eye Limits

The Eye Limits tab of this window has the following controls:

specify limits by...	Determines how the limits are specified relative to either the ideal edge location or the center of the eye.
input limits as...	Determines how the limits are specified in terms of picoseconds or by percentage of UI.
Sync Low to High Limit	Makes the low limit the same as the high limit.

When you click the Apply or OK button, eye limits are applied to the eye display, the limits template is drawn, and any failing data bars are colored red.

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Jitter Tool

Jitter Tool

Jitter Tool is a graphical jitter analysis tool for UltraPin1600, SB6G, and UltraSerial10G instruments. It provides a visual display of quantitative jitter information, allowing you to identify results of interest.

The instruments listed above take digital jitter measurements and capture binary data into capture memory (CMEM). Algorithms are then applied to the resulting binary data to produce jitter statistics such as random jitter (Rj) and deterministic data jitter (DDj). Jitter Tool displays the results in a traditional, scope-like display.

You can view data:

- In a per-edge jitter display that displays an idealized waveform with an indication of the jitter on each low-high/high-low transition.
- In an eye diagram that displays the overall jitter on all rising/falling transitions, overlaid with the pass/fail jitter limits.
- In a raw waveform display that shows the in-order waveform.

This section includes the following topics:

- Running Jitter Tool
- Main Display

- Context Display
- Jitter Analysis Display
- Tool Configuration Menu

Related Topics

- Programming SB6G Serial Jitter
- Programming UltraSerial10G Serial Jitter
- SB6G Jitter and Eye Measurement Test Programming
- UltraSerial10G Jitter and Eye Measurement Test Programming
- UltraPin1600 Time Measurement Test Programming

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